

SymPy Bug Card

Bug 63: Fourier Transform Has Wrong Sign at Negative Frequencies

Task. Compute the Fourier transform, using SymPy's convention $\int f(x)e^{-2\pi i x k} dx$, of

$$\frac{\cos x}{x^2 + 1}.$$

SymPy's answer. At $k = -1/3$, SymPy returns

$$\text{fourier_transform}\left(\frac{\cos x}{x^2 + 1}, x, k\right)\Big|_{k=-1/3} \approx -0.596972407016359.$$

Correct answer. The correct transform is

$$\frac{\pi}{2} \left(e^{-|2\pi k - 1|} + e^{-|2\pi k + 1|} \right),$$

so at $k = -1/3$ it is approximately 0.596972407016359.

Explanation. Since $\cos x = (e^{ix} + e^{-ix})/2$ and $\mathcal{F}\{1/(x^2 + 1)\}(k) = \pi e^{-2\pi|k|}$, the shift theorem gives the displayed positive, even transform. SymPy's value at $k = -1/3$ is the negative of this value, so it is not an equivalent branch representation.

Likely root cause. The diagnosis traces the bad sign to the Meijer definite integration path in `sympy/integrals/meijerint.py`: half-line Meijer formulas introduce branch-sensitive factors involving `polar_lift(k)**2`, `Abs(k)`, and a denominator `k`, which simplify to an erroneous `Abs(k)/k` factor. This belongs to a known meijerg Fourier-transform `abs/k` sign family: open issue [sympy/sympy#25137](#) documents the same artifact under `real=True`, with a maintainer root-cause comment matching this diagnosis; this integrand is plausibly a new instance.

Artifact (GitHub). [bug-report/bug-063-fourier-transform-has-wrong-sign-at-negative-frequencies/](#)